

Power BI Assignment 2 – DAX & Data Visualization

Calculated Columns:

A new column 'Category Type' was added in the Order Details table by combining the 'Category' and 'Sub-Category' columns by using “New column” and DAX, **Category_type = CONCATENATE('Order Details'[Category], " - "&'Order Details' [Sub-Category])**

The screenshot shows the DAX formula bar with the formula: `1 Category_type = CONCATENATE('Order Details'[Category], " - "&'Order Details' [Sub-Category])`. Below the formula bar, a table displays the resulting 'Category_type' values for various order details.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category_type
B-25602	561	212	3	Clothing	Saree	Clothing - Saree
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree
B-25604	157	5	9	Clothing	Saree	Clothing - Saree
B-25605	75	0	7	Clothing	Saree	Clothing - Saree
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree
B-25610	43	0	3	Clothing	Saree	Clothing - Saree
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree

Another column “Revenue per order” was created in the order details table using the DAX, **Revenue_per_order = 'Order Details'[Amount]*'Order Details'[Quantity]**

The screenshot shows the DAX formula bar with the formula: `1 Revenue_per_order = 'Order Details'[Amount]*'Order Details'[Quantity]`. Below the formula bar, a table displays the resulting 'Revenue_per_order' values for various order details.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category_type	Revenue_per_order
i02	561	212	3	Clothing	Saree	Clothing - Saree	1683
i02	119	-5	8	Clothing	Saree	Clothing - Saree	952
i03	193	-166	3	Clothing	Saree	Clothing - Saree	579
i04	157	5	9	Clothing	Saree	Clothing - Saree	1413
i05	75	0	7	Clothing	Saree	Clothing - Saree	525
i09	25	-5	4	Clothing	Saree	Clothing - Saree	100
i10	43	0	3	Clothing	Saree	Clothing - Saree	129
i11	160	-59	2	Clothing	Saree	Clothing - Saree	320
i13	1603	0	9	Clothing	Saree	Clothing - Saree	14427
i19	353	90	8	Clothing	Saree	Clothing - Saree	2824
i22	534	0	3	Clothing	Saree	Clothing - Saree	1602

Another calculated column named ‘Sales Category’ in the Order Details table was created that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value using the DAX,

Sales_category = IF('Order Details'[Amount]>AVERAGE('Order Details'[Amount]), "Above average", "Below average")

The screenshot shows the DAX formula bar with the formula: `1 Sales_category = IF('Order Details'[Amount]>AVERAGE('Order Details'[Amount]), "Above average", "Below average")`. Below the formula bar, a table displays the resulting 'Sales_category' values for various order details.

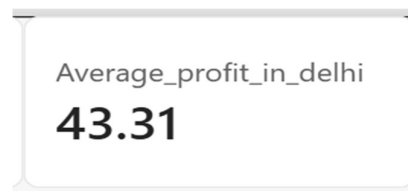
Amount	Profit	Quantity	Category	Sub-Category	Category_type	Revenue_per_order	Sales_category
561	212	3	Clothing	Saree	Clothing - Saree	1683	Above average
119	-5	8	Clothing	Saree	Clothing - Saree	952	Below average
193	-166	3	Clothing	Saree	Clothing - Saree	579	Below average
157	5	9	Clothing	Saree	Clothing - Saree	1413	Below average
75	0	7	Clothing	Saree	Clothing - Saree	525	Below average
25	-5	4	Clothing	Saree	Clothing - Saree	100	Below average
43	0	3	Clothing	Saree	Clothing - Saree	129	Below average
160	-59	2	Clothing	Saree	Clothing - Saree	320	Below average
1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above average
353	90	8	Clothing	Saree	Clothing - Saree	2824	Above average
534	0	3	Clothing	Saree	Clothing - Saree	1602	Above average
149	-87	4	Clothing	Saree	Clothing - Saree	596	Below average

Calculated Measures:

A new measure was created to count the total number of orders in the Order Details table using the DAX, **Order_count = DISTINCTCOUNT('Order Details'[Order ID])**. The order count was displayed by using the card visual.



Another measure was created to calculate the average profit for orders placed in Delhi by using the DAX, **Average_profit_in_delhi = CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[City]="Delhi")**

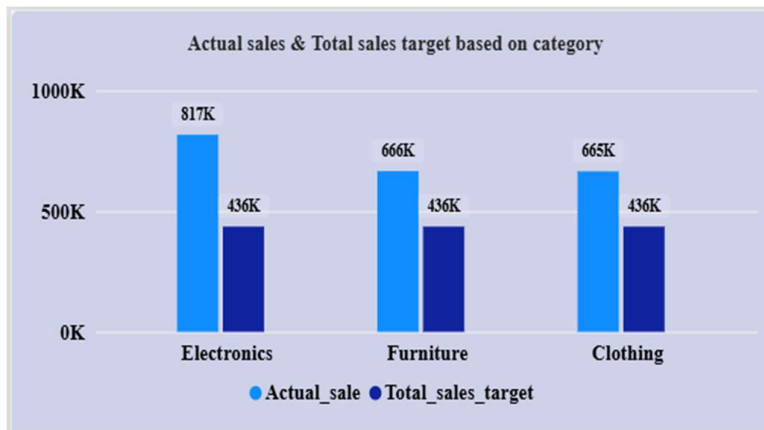


Another measure was created to calculate the total sales amount accumulated from the earliest order date up to each order date using the DAX, **Year_to_date_sale = TOTALYTD(SUM('Order Details'[Revenue_per_order]),'List of Orders'[Order Date])**. This was visualized using the table visual.

Order Date	Total_sales_amount	Year_to_date_sale
08 April 2018	₹ 14,669	₹ 51,355
09 April 2018	₹ 11,987	₹ 63,342
11 April 2018	₹ 320	₹ 63,662
12 April 2018	₹ 14,945	₹ 78,607
13 April 2018	₹ 2,172	₹ 80,779
15 April 2018	₹ 1,238	₹ 82,017
17 April 2018	₹ 1,525	₹ 83,542
18 April 2018	₹ 3,210	₹ 86,752
20 April 2018	₹ 3,041	₹ 89,793
22 April 2018	₹ 3,172	₹ 92,965
23 April 2018	₹ 7,121	₹ 1,00,086
24 April 2018	₹ 7,065	₹ 1,07,151
25 April 2018	₹ 38	₹ 1,07,189
26 April 2018	₹ 21,343	₹ 1,28,532
27 April 2018	₹ 9,615	₹ 1,38,147
28 April 2018	₹ 4,774	₹ 1,42,921
29 April 2018	₹ 9,298	₹ 1,52,219
30 April 2018	₹ 608	₹ 1,52,827
Total	₹ 21,46,870	₹ 8,36,599

Data Visualization:

Using the DAX, **Actual_sale = SUM('Order Details'[Revenue_per_order])** and **Total_sales_target = SUM('Sales target'[Total sales target])** was created. The actual sales with sales targets by category was compared using a clustered column chart.



A new calculated column profit margin was created using the DAX, **Profit_margin = ('Order Details'[Profit]/'Order Details'[Revenue_per_order])*100**

Profit_margin

type: Decimal number

Format: Decimal number

Summarization: Sum

Data category: Uncategorized

Sort by column: Sort

Data groups: Groups

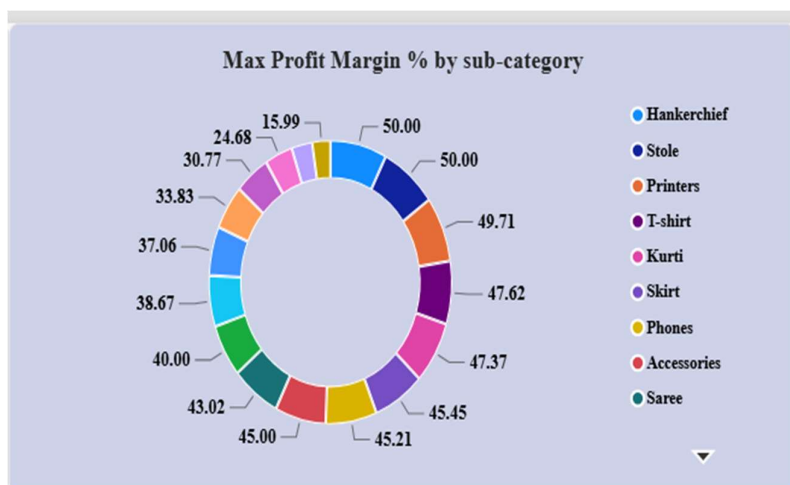
Manage relationships: Relationships

New column: Calculations

1 Profit_margin = ('Order Details'[Profit]/'Order Details'[Revenue_per_order])*100

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category_type	Revenue_per_order	Sales_category	Profit_margin
25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above average	12.60
25602	119	-5	8	Clothing	Saree	Clothing - Saree	952	Below average	-0.53
25603	193	-166	3	Clothing	Saree	Clothing - Saree	579	Below average	-28.67
25604	157	5	9	Clothing	Saree	Clothing - Saree	1413	Below average	0.35
25605	75	0	7	Clothing	Saree	Clothing - Saree	525	Below average	0.00
25609	25	-5	4	Clothing	Saree	Clothing - Saree	100	Below average	-5.00
25610	43	0	3	Clothing	Saree	Clothing - Saree	129	Below average	0.00
25611	160	-59	2	Clothing	Saree	Clothing - Saree	320	Below average	-18.44
25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above average	0.00
25619	353	90	8	Clothing	Saree	Clothing - Saree	2824	Above average	3.19
25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above average	0.00
25623	149	-87	4	Clothing	Saree	Clothing - Saree	596	Below average	-14.60
25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above average	-10.99
25628	24	-9	4	Clothing	Saree	Clothing - Saree	96	Below average	-9.38
25633	711	-8	4	Clothing	Saree	Clothing - Saree	2844	Above average	-0.28
25635	382	30	3	Clothing	Saree	Clothing - Saree	1146	Above average	2.62

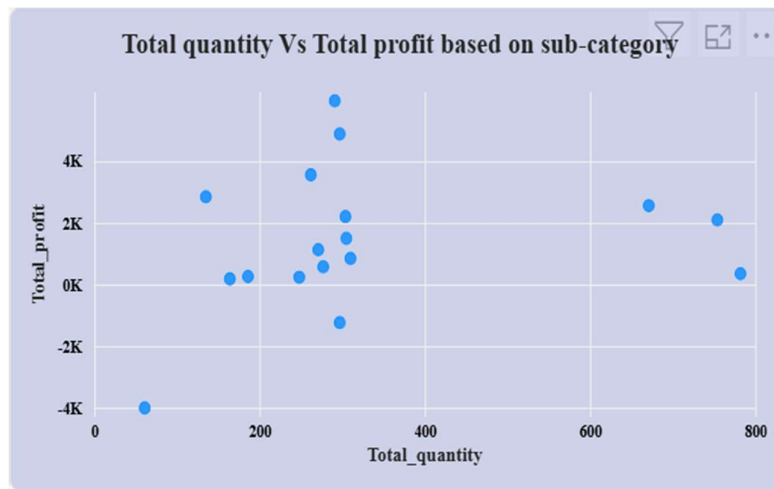
Then the new measure maximum profit margin was created using the DAX, **Max Profit Margin % = MAX('Order Details'[Profit_margin])**. Then the maximum profit margin for each sub-category of products was visualized using a donut chart.



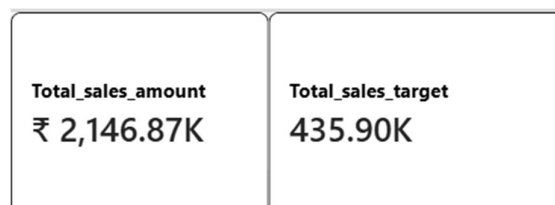
The trend of monthly sales over time was visualized using a line chart.

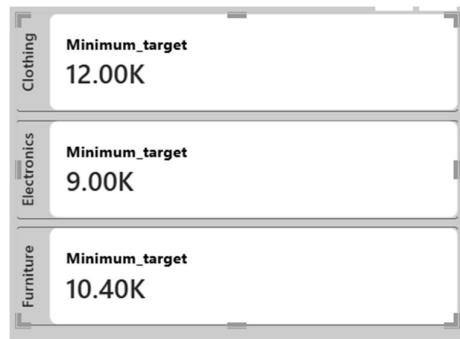


The relationship between profit and quantity sold for different sub-categories was compared using a scatter chart.



Card visuals was created using the DAX, **Total_sales_amount** = SUM('Order Details'[Revenue_per_order]), **Total_sales_target** = SUM('Sales target'[Total sales target]) to display the total sales amount alongside the sales target for quick comparison and analysis. A multi-row card was created to display the minimum target for each segment.

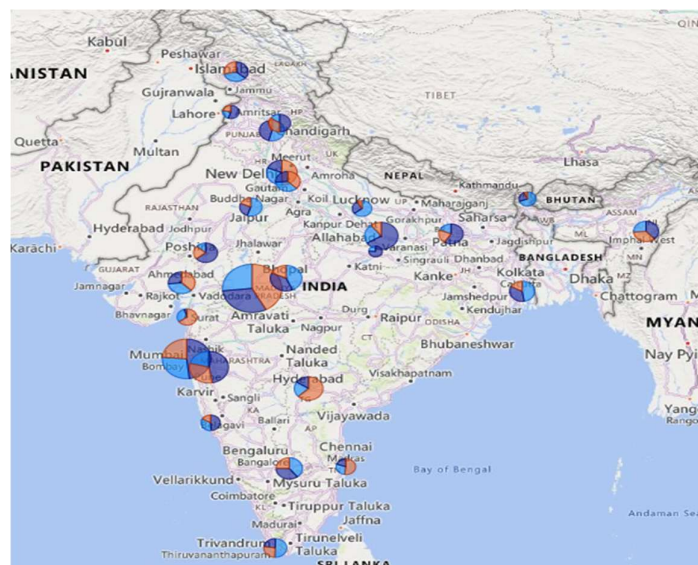




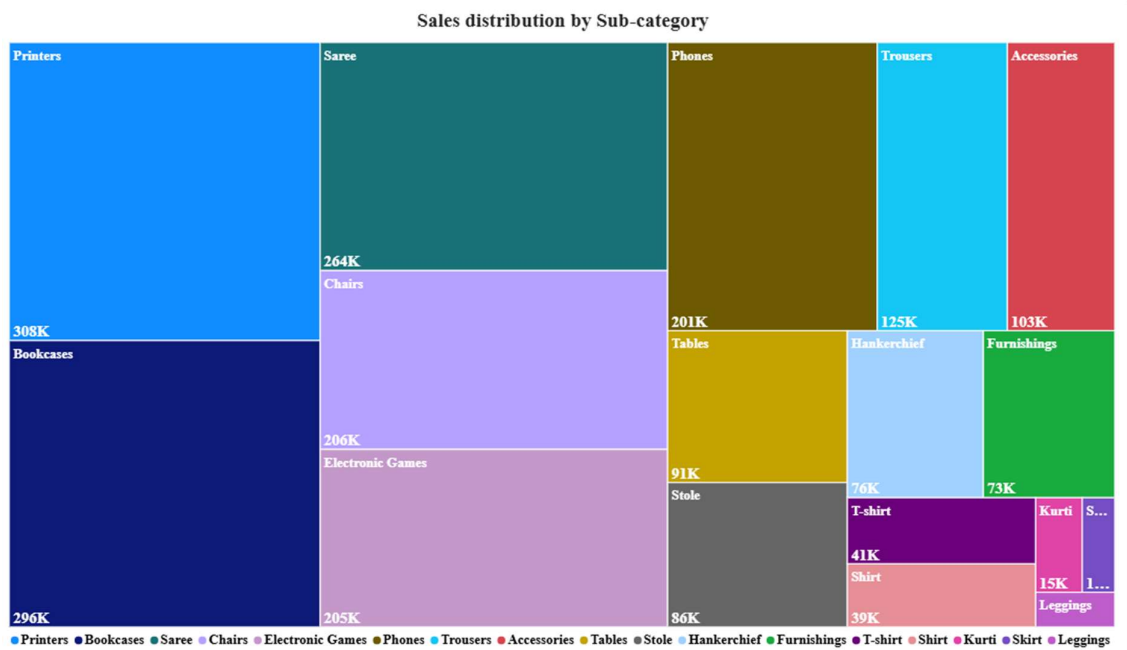
A matrix view was created to analyze the actual sales compared to sales targets across different categories and months.

Category	Clothing		Electronics		Furniture		Total	
Month	Actual_sale	Total_sales_target	Actual_sale	Total_sales_target	Actual_sale	Total_sales_target	Actual_sale	Total_sales_target
January	66935	435900	161965	435900	108329	435900	337229	435900
February	49332	435900	48507	435900	79781	435900	177620	435900
March	92718	435900	98989	435900	130043	435900	321750	435900
April	67649	435900	43637	435900	41541	435900	152827	435900
May	44338	435900	61765	435900	32664	435900	138767	435900
June	40292	435900	43234	435900	26489	435900	110015	435900
July	13060	435900	22939	435900	17145	435900	53144	435900
August	59474	435900	47050	435900	44990	435900	151514	435900
September	58752	435900	33543	435900	40422	435900	132717	435900
October	55979	435900	69151	435900	51533	435900	176663	435900
November	72687	435900	92824	435900	64274	435900	229785	435900
December	43306	435900	92979	435900	28554	435900	164839	435900
Total	664522	435900	816583	435900	665765	435900	2146870	435900

Geographic Sales Analysis was created based on the total sales by city using the map visual to identify regional sales patterns. To get a perfect location, new calculated column Map location was created using the DAX, **Map_Location = CONCATENATE('List of Orders'[City]," - " & 'List of Orders'[State]&" - India")**



The sales distribution across different sub-categories was represented using a treemap.



A funnel chart was created to visualize the distribution of order counts across different states.

